Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/15/2020
PI Name:	Pantalos, George Ph.D.		
Project Title:	Creating Surgical Capabilities for Exploration Space Flight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	40202-1572	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO NNJ16ZSA001N-TRIRT. Appendix C: Translational Research Institute for Space Health (TRISH) Research Topics
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Badger, Julia Ph.D. (NASA Johnson Space Center) Broderick, Timothy M.D. (Wright State University) Grant, Gerald D.M.D. (University of Louisville School of Dentistry) Miller, Karen Ph.D. (University of Louisville School of Medicine) Sharp, Keith Sc.D. (University of Louisville) Sutton, Erica M.D. (University of Louisville) Wu, Jimmy B.S. (Baylor College of Medicine)		
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Task Description:	Because of the difficulty of resupplying from Earth and the long communication delay during exploration space missions, the crew must be self-supported, including for healthcare. For long-duration flights, it is possible and perhaps even probable, that surgical procedures may be required to treat trauma or disease. Astronauts could make use of stowed surgical instruments as well as those rapidly created using 3D printing technology, to perform potentially life-saving emergency procedures. With small crews, rapid training programs and expert robotic surgical assistants would prove very helpful. To supplement pre-flight training, crew medical officers could be educated in-flight on specific surgical skills as needs arise. This project will develop just-in-time surgical training, optimize the supply logistics of manifested versus 3D printed surgical instruments, and modify the NASA Robonaut 2 to add responsive movements for assistance during surgery. This project focuses on an example surgical procedure (appendicitis), preliminary to consideration of broader surgical needs. Training material utilizing virtual or augmented reality will be developed and evaluated for effectiveness. The NASA Robonaut 2 human-like robot will be modified to respond to voice commands and deliver necessary tools. Its performance will be evaluated for efficacy and benefits in aiding crew medical officers. To maximize the space available for other equipment, surgical supplies, materials, and instruments needed for treatment of the appendicitis example will be analyzed to identify what should be stowed before launch and what would be more appropriately rapidly 3D printed in-flight as needed. By incorporating a structured training regimen available for in-flight review, with rapid 3D printing capabilities for surgical instruments and with a responsive robotic assistant, surgical procedures during exploration space flight could be safely and effectively performed. These capabilities will enhance not only crew welfare, but also the success of the mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	FINAL REPORTING SUBMITTED DECEMBER 2019 SPECIFIC AIM 1 Impact of Key Findings: The major finding is that is it possible to do a comprehensive multi-performer task analysis for a laparoscopic appendectomy (and by extension many other medical procedures) that can be replicated by performers with basic knowledge. This process also applies to future revisions of the job aid that will incorporate updated surgical and anesthesia methods, technology, and place the procedure in the context of an exploration spacecraft. A secondary finding is the identification of steps in this medical procedure that need further research, which are generalizable skills (i.e., that instructional segment has multiple applications), and which items of supply and equipment are multi-purpose. Additional findings include the design of a multi-performer task analysis spreadsheet suitable for continuous revision to accommodate new findings, and that simplification of the procedure can reduce the items of equipment needed from more than 50 to fewer than 30. SPECIFIC AIM 2 Impact of Key Findings: Successful conversion of conventional surgical instruments into digital CAD (computer assisted design) files that could be fabricated as-needed, on demand opens a new operational paradigm where instruments and parts do not need to be manifested from Earth. This provides considerable savings in vehicle mass and volume for equipment and parts that are only on the exploration-class vehicle as a just-in-case. Considering that exploration-class missions cannot be resupplied from Earth, the mass-less and volume-less digital CAD file can be turned into a necessary part or instrument as-needed thereby optimizing the limited stowage mass/volume in the vehicle. The process for quantifying the seven operational tenants enables evaluation, management, and trade-offs to be made on the optimal logistics/resupply methodology for an exploration-class mission. Though our study did not provide an optimal logistics/resupply model, a manager could weigh the seven operational tenants to determine the superior logistics/resupply model based on mission need and constraints. SPECIFIC AIM 3 Impact of Key Findings: Tests demonstrated a dexterous robot was typically able to retrieve requested tools within its workspace. However, the current time for visual processing and motion planning calculations required by the robot to achieve this complex task was prohibitive for proper team resource management in a real-time surgical situation. Though no algorithmic optimization was attempted, since this was beyond this grant's scope, future human-robot collaboration systems must remain cognizant of potential workload increases caused by having to pre-determine task steps and assign robots to compensate for slow task execution speeds. The robotic surgical assistance demonstration achieved incremental success. Difficulties with the distal force-torque sensors in the robot's forearms led to hybrid force-position control being positionally inaccurate, so it was abandoned in favor of torque limited position control. While the robot achieved probe placement, the robot's workspace and constrained motion planner limited the number of ultrasound locations and orientations. Three ultrasound locations were tested each with six possible angles leading to a total of eighteen unique ultrasound positions. The outcomes of the Blue Sky conferences are available as a Summary Report. An expanded version of the Summary Report that included all of the presentation slides and a bibliography was published as a NASA Technical Publication: NASA/TP-2018-220341. Several of the conclusions resulting from the symposium were adopted by the Specific Aim 1 team as assumptions to provide context. For example, the composition and medical training level described at Blue Sky (a crew of 6 including 1 physician and 5 with the training equivalence to an EMT) provided the knowledge/skill level for their surgical procedure task analysis. REPORTING SUBMITTED JANUARY 2019 This research project is creating some of the procedures and instruments that will be necessary to perform a surgical procedure during an exploration space flight. Because the crew will be small, we are also evaluating if a human-inspired, dexterous robot (e.g., Robonaut 2) can be useful as a surgical/medical assist to help with this procedure. This undertaking is more challenging because we are also assuming that the crew medical officer (CMO) of the mission will have a medical background, but may not necessarily be trained as a surgeon. Given this scenario, the "job aid" (360 degree virtual reality video) that will be developed to guide the CMO through the surgical procedure may establish the method to create job guides for other medical and non-medical procedures and give us a better understanding of what it would take to consider doing a surgical procedure in such remote and extreme circumstances. Modifying a robot so that it can respond to the verbal commands of a surgeon to assist in a surgical procedure paves the way for responsive robot activity of many kinds on an exploration spacecraft as well as helping to define the characteristics of tools and instruments that a dexterous robot can successfully identify, grasp, manipulate, and hand off. As all of these capabilities are being developed and tested, a repetitive logistical analysis will be conducted to assess the optimal balance between what material, supplies, instruments, and equipment for a minimally invasive surgical case need to be loaded onto the spacecraft prior to launch verses created during the mission if ever needed.

Task Progress:

FINAL REPORTING SUBMITTED DECEMBER 2019
AIM 1 Original Project Aims/Objectives: Specific Aim 1 Objectives were to create and evaluate the efficacy of audio-visual presentation platforms to provide just-in-time support of surgical tasks for non-surgeons and to consider the impact of non-surgeons performing a procedure during exploration space flight. We researched which variation of the procedure (laparoscopic appendectomy) was best for replication by non-surgeons, which supplies and equipment needed modification, and ways to task analyze actions, supplies, and equipment.

Key findings: A searchable task analysis spreadsheet in Excel can house all needed data in a format and can be converted into a digital data base. Video formats that can capture the task steps are a 4 camera view video, and a 3600 virtual reality (VR) video of the entire procedure.

Impact of Key Findings: Steps in the procedure can be marked for: further research, generalizable skills, and what supplies and equipment are multi-purpose. VR video can capture all steps of a laparoscopic appendectomy to support replication by performers with basic medical knowledge.

Proposed Research for the future: 1) The current laparoscopic tower is too large and must be miniaturized for space flight; other approaches need to be considered. 2) Anesthetic and other pharmaceuticals must be researched. 3) The next version of the VR should be produced in an accurate mockup of the space craft module with a robot assistant - an augmented reality (AR) job aid may be more useful.

AIM 2 Original Project Aims/Objectives: Specific Aim 2 objectives were to create computer aided design (CAD) models of surgical instruments and to assess the operational resource footprint of providing instruments as-needed. Three logistics/resupply models were considered: 1) Fully manifested from Earth with conventional surgical instruments. 2) Partially manifested from Earth with 3-D printed surgical instruments to resupply depleted surgical instruments printed as-needed. 3) Fully 3-D printed surgical instruments fabricated as-needed. Models were assessed on 7 operational tenants: mass, energy, volume, time, cost, risk, and crew competence, providing a strategy to evaluate trade-offs between the models to determine the optimal plan for providing surgical instruments.

Key Findings: A process for converting selected conventional surgical instruments into CAD models and 3-D printing was implemented for 27 different instrument designs. A process for assessing the 7 operational tenants and quantifying those values for the exploration-class surgical kit was implemented.

Impact of Key Findings: Most conventional instruments can be converted into CAD files to print instruments as-needed thus adding a new operational paradigm where instruments don't need to be manifested from Earth. A process for quantifying the 7 operational tenants including evaluation, management, and trade-offs allows managers to determine the superior logistics/resupply model based on mission need and constraints.

Proposed Research for the Future: 1) Conduct a validation study of 3-D printed surgical instruments printed in 1-G, and on the International Space Station (ISS) in micro-G using the Made in Space 3-D printer. 2) Assess partially 3-D printed instruments to determine feasibility and threshold for when an instrument should be considered for 3-D printing. 3) Conduct the logistics/resupply analysis with recycling capability to determine the superior model (Note: recycling source material for 3D printing on the ISS will be available in the near future).

AIM 3 Original Project Aims/Objectives: Aim 3 objectives were to research and create the surgical assistant robot's initial ability to identify, grasp, and hand off surgical tools following a verbal request, to perform an ultrasound on a CPR training torso following verbal commands, and to conduct a Blue Sky Symposium to explore the role of robotic involvement in surgery during space flight.

Key Findings: Robonaut 2 (R2) was able to provide surgical assistance with design refinements. The scalpel handle was thickened, colored, and had an ergonomic curve to aid in the visual identification. Eliminating shiny surfaces aided the identification of instruments, as well as a solid color plane background. An angled platform for instrument staging also improved performance. Programming design refinements were needed for R2 to hold an ultrasound imaging probe at the correct location, angle, and contact pressure. Unlike the surgical instruments which were adjusted to help visual detection and robotic grasping, the ultrasound probe was an unmodified clinical probe. The Blue Sky Symposium, "Minimally Invasive Expeditionary Surgical Care using Human-Inspired Robots," included 22 participants from robotic surgical companies, surgeons, academic surgical robotics developers, and a physician/astronaut.

Impact of Key Findings: R2 was typically able to retrieve requested instruments within its workspace. However, the current time for visual processing and motion planning calculations were prohibitive for proper team resource management in a real-time surgical situation. The robotic surgical assistance demonstration achieved incremental success. Difficulties with the distal force-torque sensors in the robot's forearms led to hybrid force-position control being positionally inaccurate, so it was abandoned in favor of torque limited position control. The outcomes of the Blue Sky conference are available as a Summary Report. An expanded version including the presentation slides and a bibliography was published as NASA Technical Publication: NASA/TP-2018-220341.

Proposed Research for the Future: 1) Continue to refine R2's dexterity, range of motion, and ability to recognize and hand off surgical instruments, and respond to verbal commands in the surgical setting by incorporating an improved vision system and faster controller processing speed. 2) Support further interdisciplinary symposiums to share expertise.

REPORTING SUBMITTED JANUARY 2019

(1) Original project specific aims/objectives:

a) Determine efficacy of visual/audio presentation platforms in providing just-in-time and refresher training of surgical tasks for non-surgeons and surgeons. Various content presentation platforms will be evaluated in their effectiveness in providing rich training content to clinical and non-clinical operators executing surgical tasks. This will determine the best training tools for instilling and retaining surgical skills.

b) Quantify the operational resource trade-off between conventionally manifested and those 3-D printed as-needed, fabricated surgical equipment. The operational resources of mass, volume, crew time, etc. needed to support space flight surgical capabilities and hardware will be characterized to understand the trade-offs between the various resupply/logistics methodologies that could be used on exploration-class space flight missions.

c) Determine efficacy of human-inspired robotic operators as assistants in space flight surgeries. Performance evaluations will be made with human-inspired robots using surgical tools, receiving commands from a human operator,

	and performing surgical tasks to evaluate their effectiveness as surgical assistants. (2) Key findings: Specific Aim 1: The task analysis process is a useful tool to comprehensively determine what is required to conduct a minimally invasive surgical procedure such as an appendectomy in the extreme environment of exploration space flight. A surgeon working in a fully staffed surgical facility performs the most critical steps in the procedure, but is supported by other highly trained staff who also perform essential tasks. A task analysis identifies and describes all actions taken by all staff members to work toward a successful outcome. When developed as a job aid, it serves as a detailed guide to replicating the procedure, even in extreme conditions. Specific Aim 2: It is certainly possible to create several kinds of useful and acceptable surgical instruments by additive manufacturing (3-D printing) for use by human crew members and dexterous robots. However, 3-D printing is a very involved process that takes quantities of time and source material, which may limit extensive use to support surgical procedures in exploration space flight. This is especially true when you consider the number of instruments needed, the printing yield rate, and that some of the instruments are so complex (e.g., stapler/scalpel, surgical clip device) that it would not be reasonable to create them by 3-D printing in any environment. These more complex instruments would need to be conventionally manifested in sterile packaging prior to launch. Specific Aim 3: A human inspired, dexterous robot with proper programing, sensors, and instrumentation/effectors is capable of responding to a verbal command, independently identifying a surgical instrument, grasping the instrument, manipulating the instrument, and passing it off to the person delivering the command. The design of the instrument may need to be modified to increase the accuracy of correct identification and improving the ability to grasp and manipulate it. 3-D printing has been a useful method for taking modified instrument designs and creating instruments of the new design in rapid succession for evaluation with a human-inspired, dexterous robot. Current robot processing time is too slow for these tasks to be performed in real-time, but increased processing capability (which is beyond the scope of this grant) could result in acceptable robot response times. (3) Impact of key findings: At this intermediate stage of development, it is not possible to determine the efficacy of the virtual reality "job guide." The detailed task analysis process is a robust method to get the job guide in the best possible readiness to be an effective tool for this application. Attractive as the ability to create 3-D printed surgical instruments is, continued critical logistical evaluation is necessary to clarify the value, reasonableness, and role of 3-D printed instruments for surgical procedures in exploration space missions. From the perspective of crew resource management, there is great value in being able to substitute a human crew member with a human-inspired, dexterous robot as a surgical assistant. However, the current slower robotic processing time needs to be improved for a dexterous robot to be useful in this application. (4) Current Results: Intraoperative imaging of minimally invasive appendectomy procedures have helped to generate a detail task analysis and storyboard. The storyboard is the basis for the production of a virtual reality job guide for the crew member who needs to perform an appendectomy or related procedure during an exploration space mission. It is possible to create a variety of functional, 3-D printed surgical instruments appropriate for use by both a human crew member and a human-inspired, dexterous robot. The overall logistical benefit of this approach will be analyzed in Year 2 of the project. The use of a human-inspired, dexterous robot as a surgical assistant is reasonable to consider if the limitation of slow task processing time can be rectified.
Bibliography Type:	Description: (Last Updated: 07/24/2025)
Abstracts for Journals and Proceedings	Pantalos, G, Sutton E, Sharp K, Miller K, Grant G, Broderick T, Wu J, Badger J, Doerr H. "Creating Surgical Capabilities for Exploration Space Flight." 2018 NASA Human Research Program Annual Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Annual Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Articles in Peer-reviewed Journals	Pantalos GM, Heidel JS, Jain IM, Warner SE, Barefoot TL, Baker RO, Hailey M. "Intravenous fluid resuscitation capabilities in simulated reduced gravity." Aerosp Med Hum Perform. 2023 Aug;94(8):596-603. https://doi.org/10.3357/AMHP.6151.2023 ; PMID: 37501295 , Aug-2023
Articles in Peer-reviewed Journals	Miller KH, Sutton E, Pantalos G. "Hierarchical task analysis reimagined as a planning tool for surgery during exploration space flight." Surg Innov. 2022 Oct;29(5):616-624. https://doi.org/10.1177/155335062110532 ; PMID: 34865575 , Oct-2022
Significant Media Coverage	Pantalos G. "George Pantalos guest on the National Public Radio show Science Friday, "How Andy Weir Engineered A Lunar City In 'Artemis.' " National Public Radio show Science Friday, November 17, 2017. https://www.sciencefriday.com/segments/how-andy-weir-engineered-a-lunar-city-in-artemis/ ; accessed 1/4/2019., Nov-2017